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Knee kinematics and joint moments during gait following anterior cruciate ligament reconstruction: a systematic review and meta-analysis

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ABSTRACT

Background Abnormal gait after anterior cruciate ligament reconstruction (ACLR) may contribute to development and/or progression of knee osteoarthritis.
Objective To conduct a systematic review and meta-analysis of knee kinematics and joint moments during walking after ACLR.
Methods We searched seven electronic databases and reference lists of relevant papers, for cross-sectional, human-based observational studies comparing knee joint kinematics and moments during level walking in individuals with ACLR, with the uninjured contralateral knee or healthy individuals as a control. Two independent reviewers appraised methodological quality (modified Downs and Black scale). Where possible, data were pooled by time post-ACLR (RevMan), otherwise narrative synthesis was undertaken.
Results Thirty-four studies were included. Meta-analysis revealed significant sagittal plane deficits in ACLR knees. We found greater knee flexion angles (standardised mean difference: 1.06; 95% CI 0.39 to 1.74) and joint moments (1.61; 0.87 to 2.35) <6 months post-ACLR, compared to healthy controls. However, lower peak knee flexion angles were identified 1–3 years (−2.21; −3.16 to −1.26) and ≥3 years post-ACLR (−1.38, −2.14 to −0.62), and lower knee flexion moment 6–12 months post-ACLR (−0.76; −1.40 to −0.12). Pooled data provided strong evidence of no difference in peak knee adduction moment >3 years after ACLR (vs healthy controls) (0.09; −0.63 to 0.81). No transverse plane conclusions could be drawn.
Conclusions Sagittal plane biomechanics, rather than the knee adduction moment, appear to be more relevant post-ACLR. Better understanding of sagittal plane biomechanics is necessary for optimal post-operative recovery, and to potentially prevent early onset and progression of knee OA after ACLR.
Trial registration number PROSPERO systematic review protocol registration number CRD4201400882 2.

INTRODUCTION

Post-traumatic knee osteoarthritis (OA) has been reported in over 50% of individuals within 10–20 years after anterior cruciate ligament reconstruction (ACLR).[1] Multiple factors contribute to initiation of post-traumatic knee OA, including intra-articular pathogenic processes initiated at the time of injury, as well as mechanical factors. It is thought that abnormal knee kinematics after ACLR contribute to the initiation of degenerative processes, due to changes in cartilage loading.[2–4] Since not all individuals develop knee OA after ACLR; it is important to understand whether variations in mechanical factors may be a risk factor in those individuals who develop knee OA after ACLR.

To date, studies investigating gait characteristics following ACLR provide inconsistent results regarding knee kinematics and joint moments. This may reflect between-study variability in patient population characteristics (eg, graft type, sex, time post-ACLR), comparator (healthy controls vs contralateral knees), data acquisition and processing, variables of interest (eg, mean or peak values, range of motion) and data presentation (eg, normalisation). Thus, the need to synthesise comparable knee kinematics and joint moment data is evident. Identifying potentially modifiable gait characteristics in ACLR knees may facilitate the development of specific interventions to prevent early onset and progression of post-traumatic knee OA.

Three systematic reviews have explored gait patterns in individuals who had undergone ACLR.[5–7] Shi *et al*,[5] conducted meta-analyses on knee sagittal plane kinematics and moments and transverse plane kinematics during level walking, while, Hart *et al*,[6] and Gokeler *et al*,[7] qualitatively reported knee kinematics and moments without conducting meta-analyses. Collectively, these reviews reported the presence of gait alterations in the sagittal, frontal and transverse planes after ACLR. However, since two of these reviews did not conduct meta-analyses,[6 7] and additional studies have been published since the last meta-analysis was published in 2010,[5] an updated systematic review with meta-analysis on knee kinematics and joint moments associated with ACLR is warranted.

Altered biomechanics are often reported in individuals with ACLR knees during high-demand activities such as hopping and vertical jump.[8 9] However, these activities are often limited to individuals returning to sport. Level walking is a task that all individuals participate in, and is frequently investigated in non-traumatic knee OA populations. Therefore, it is important to identify knee kinematics and moments during level walking after ACLR to understand its potential implications for post-traumatic knee OA. The aim of this systematic review and meta-analysis was to determine knee kinematics and joint moments in the sagittal, frontal and transverse planes during walking in ACLR knees, compared to healthy controls and unaffected contralateral knees.

METHODS

The study protocol was developed in consultation with guidelines provided by the PRISMA (Preferred

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Reporting Items for Systematic Reviews and Meta-Analyses) statement.[10] The protocol was prospectively registered on the PROSPERO International register for systematic reviews (http://www.crd.york.ac.uk/PROSPERO) (CRD4201400882 2).

Literature search strategy

Using guidelines provided by the Cochrane Collaboration, a comprehensive search strategy was devised from the following electronic databases with no date restrictions: (i) MEDLINE via OVID; (ii) EMBASE via OVID; (iii) CINAHL via EBSCO; (iv) Scopus; (v) Web of Science; and (vi) SPORTDiscus. The MEDLINE search strategy was adapted for the other databases (see online supplementary material A). Keywords included 'anterior cruciate ligament', 'ACL', 'ACL (surg* or reconstruct*), 'biomechanic', 'moment', 'angle*', 'torque', 'rotation', 'kinetic*', 'kinematic*', 'joint load'. The search was limited to English language and full-text articles. Two teams of two investigators (total of 4: HFH and DCA; AGC and SMC) reviewed all titles returned by the database searches, and retrieved suitable abstracts. Where abstracts suggested that papers were potentially suitable, the full-text versions were retrieved and included in the review if they were found to fulfil the selection criteria. Reference lists of all publications considered for inclusion were hand-searched recursively until no additional eligible publications were identified.

Selection criteria

Selection criteria were as follows: cross-sectional (ie, compares different population groups at defined point), human observational studies comparing level walking knee joint biomechanics of individuals with ACLR, with either the uninjured contralateral knee or healthy controls as a comparator. No restriction was placed on type of graft, sex, age, time since surgery or recruitment method. For studies investigating effects of interventions such as knee bracing on knee joint biomechanics, baseline data were included. Studies permitting participants to walk with walking aids were excluded.

Assessment of methodological quality and risk of bias

The methodological quality of included studies was rated using a modified version of the Downs and Black checklist.[11] The modified version had a maximum score of 15 (see online supplementary material B), where a total score of ≥12 indicates high methodological quality, a score of 10 or 11 indicates moderate quality, and a score ≤9 indicates low quality.[12] Two independent reviewers (AGC and ZM), who remained blind to authors, affiliations, and the publishing journal, rated each study on the 15 item criteria. Any inter-rater disagreement was discussed in a consensus meeting, and unresolved items were taken to a third reviewer (NJC) for consensus.

Data management and statistical analysis

Data pertaining to study characteristics (eg, graft type, time since surgery), knee kinematics and joint moments (described as external moments) were independently extracted and entered into an Excel spreadsheet. If sufficient data were not reported in the published article or supplementary material provided, the corresponding author was contacted to request further data. Data were analysed based on time post-ACLR: (1) less than 6 months (6 months); (2) 6 months to less than 12 months (6–12 months); (3) 12 months to less 3 years (1–3 years); and (4) 3 years and over (≥3 years). Methodological heterogeneity was evaluated for all outcome measures (eg, peak knee flexion angle, peak knee flexion moment, peak knee adduction angle, peak knee adduction moment, peak knee internal rotation angle, peak knee internal rotation moment) and data were pooled for meta-analysis (Review Manager V.5.3) where possible. Moment data are expressed as external, unless otherwise stated. Standardised mean differences (SMD) with 95% CI were calculated for variables of interest and random effects models were used for each meta-analysis. The magnitude of the pooled SMD was interpreted based on Cohen's criteria, where SMD ≥0.8 indicated larger effect, moderate if >0.5 and <0.8, and weak if 0.2–0.5.[13] Where possible, sensitivity analyses were

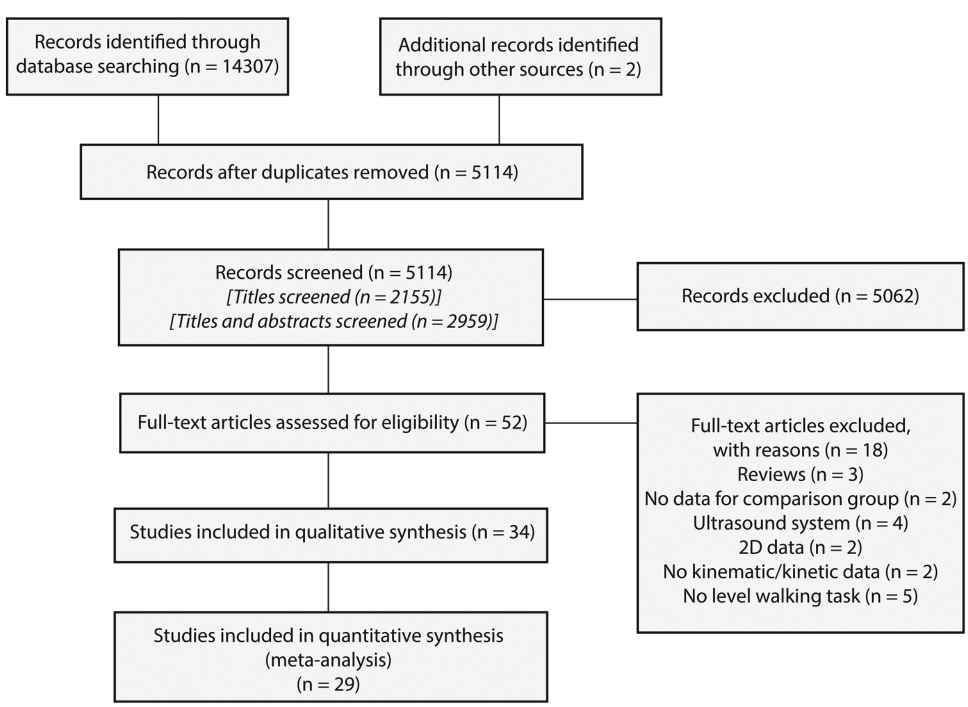

Figure 1 Flow chart of the study selection process.

Table 1 Details of included studies

Author, year	ACLR cohort	Graft type*	Time post-ACLR	Comparator cohort
Bacchini *et al*, 2009[57]	n=8, male/female 8/0, age 28 (21–33) Y	n=8 bone patellar tendon bone autograft	6 (5–7) M	Contralateral knee
Bulgheroni *et al*, 1997[58]	n=15, male/female 15/0, age 25±3 Y	n=15 bone patellar tendon bone autograft	17±5 M	n=5, male/female 5/0, age 28 ±3 Y
Bush-Joseph *et al*, 2001[33]	n=22, male/female 13/9, age 27±11 Y	n=22 patellar tendon graft	22±12 M	n=22, male/female 13/9, age 29±8 Y
Butler *et al*, 2009[39]	n=17, male/female 4/13, age 24±6 Y, BMI 24±3 kg/m^{-2}	–	5±4 Y	n=17, male/female 4/13, age 23±6 Y, BMI 23±2 kg/m^{-2}
Decker *et al*, 2004[21]	Group 1 (ACLR): n=8, age 26±7 Y, height 1.8 ±0.1 m, weight 75±8 kg Group 2 (ACLR): n=8, age 30±7 Y, height 1.7 ±0.1 m, weight 76±15 kg	n=8 patellar tendon graft	6 W	n=8, age 28±4 Y, height 1.70 ±0.10 m, weight 72±11 kg
Devita *et al*, 1997[22]	n=9, male/female 6/3, age 22 Y	n=9 patellar tendon graft	3 W, 5 W	n=10, male/female 5/5, age 21 Y
Devita *et al*, 1998[59]	n=8, male/female 5/3, age 21 Y, weight 74 kg	n=8 patellar tendon graft	3 W, 6 M	n=22, male/female 14/8, age 22 Y, weight 74 kg
Di Stasi *et al*, 2013[24]	Group 1 (RTS-Fail): n=22, male/female 14/8, age 31 ±12 Y Group 2 (RTS-Pass): n=20, male/female 16/4, age 25±10 Y	n=44 soft tissue allograft or semitendinosus/ gracilis autograft	6 M	Contralateral knee
Ferber *et al*, 2002[15]	n=10, male/female 5/5, age 28±9 Y, height 1.7 ±0.2 m, weight 79±14 kg	n=10 patellar tendon graft	3 M	n=10, male/female 5/5, age 24 ±3 Y, height 1.7±0.1 m, weight 67±11 kg
Ferber *et al* 2003[16]	n=10, male/female 5/5, age 28±9 Y, height 1.7 ±0.2 m, weight 79±14 kg	n=10 patellar tendon graft	3 M	n=10, male/female 5/5, age 25 ±3 Y, height 1.7±0.1 m, weight 77±11 kg
Ferber *et al*, 2004[29]	n=10, male/female 5/5, age 28±9 Y, height 1.7 ±0.2 m, weight 79±14 kg	n=10 patellar tendon graft	6±5 Y	n=10, male/female 5/5, age 24 ±3 Y, height 1.7±0.1 m, weight 67±11 kg
Gao *et al*, and Zheng 2010[19]	n=14, male/female 12/2, age 25±6 Y, height 1.8 ±0.1 m, weight 83±15 kg	n=7 hamstring tendon graft, 5 patellar tendon graft, 2 Achilles tendon graft	12 M	n=15, male/female 12/3, age 23±3 Y, height 1.8±0.1 m, weight 77±16 kg
Georgoulis *et al*, 2003[17]	n=21, male/female 19/2, age 25±4 Y, height 1.7 ±0.1 m, weight 69±8 kg	n=21 patellar tendon graft	30±17 W	n=10, male/female 8/2, age 25 ±4 Y, height 1.7±0.1 m, weight 62±12 kg
Hall *et al*, 2012[34]	n=15, male/female 7/8, age 26±4 Y, height 1.7 ±0.1 m, weight 75±15 kg	n=6 hamstring tendon graft, 6 patellar tendon graft, 2 combined cadaver and hamstring tendon graft, 1 cadaver graft	6 (2–18) Y	n=17, male/female 7/10, age 26±4 Y, height 1.7±0.1 m, weight 68±12 kg
Hooper *et al*, 2002[35]	Group 1 (6 M post-ACLR): n=8, male/female 5/3, age 33±11 Y, height 1.7±0.1 m, weight 77±15 kg Group 2 (12 M post-ACLR): n=9, male/female 4/5, age 34±11 Y, height 1.7±0.1 m, weight 77±16 kg	Group 1: n=8 patellar tendon graftGroup 2: n=9 patellar tendon graft	Group 1: 6±1 M Group 2: 12±1 M	Contralateral knee
Hunt *et al*, 2010[60]	n=12, male/female 5/7, age 21±5 Y, height 1.7 ±0.1 m, weight 70±15 kg	n=12 hamstring tendon autograft	31±12 D	Contralateral knee
Karimi *et al*, 2013[30]	n=15, age 33±3 Y, height 1.8±0.1 m	n=15 hamstring or peroneus longus tendon graft	6 M	n=15, age 32±3 Y, height 1.7 ±0.1 m
Noehren *et al*, 2013[23]	n=20, male/female 0/20, age 25±6 Y, height 1.6 ±0.1 m, weight 64±8 kg	n=8 hamstring tendon graft, 7 patellar tendon graft, 5 autograft	5±3 Y	n=20, male/female 0/20, age 26±5 Y, height 1.6±0.1 m, weight 61±5 kg
Patterson *et al*, 2014a[40]	n=14 (17 limbs), male/female 0/14, age 24±3 Y, height 1.6±0.1 m, weight 65±9 kg, BMI 24±2 kg/m^{-2}	n=8 hamstring tendon graft, 9 patellar tendon graft	4±3 Y	n=17, male/female 0/17, age 21±1 Y, height 1.7±0.1 m, weight 65±7 kg, BMI 24±1 kg/m^{-2}
Patterson *et al*, 2014[20]	n=14 (17 limbs), male/female 0/14, age 24±3 Y, height 1.6±0.1 m, weight 65±9 kg	n=8 hamstring tendon graft, 9 patellar tendon graft	4±3 Y	n=17, M/F 0/17, age 21±1 Y, height 1.7±0.1 m, weight 65 ±7 kg
Roewer *et al*, 2011[26]	n=26, male/female 18/8, age 30±11 Y, height 1.8 ±0.1 m, weight 90±18 kg	n=26 semitendinosus-gracilis autograft or soft-tissue allograft	6 M, 2 Y	Contralateral knee
Sanford *et al*, 2012[41]	n=10, male/female 4/6, age 28±7 Y, height 1.7 ±0.1 m, weight 78±20 kg	n=1 hamstring tendon graft, 9 patellar tendon graft	7±6 Y	n=12, male/female 6/6, age 25 ±4 Y, height 1.7±0.1 m, weight 69±14 kg
Sato *et al*, 2013[43]	n=7, male/female 3/4, age 21±5 Y	n=7 hamstring tendon graft	14±2 M	Contralateral knee
Scanlan *et al*, 2010[28]	n=26, male/female 11/15, age 31 Y, height 1.7 m, weight 68 kg	n=12 Achilles tendon, patellar tendon, soft-tissue allograft, 14 hamstring tendon or patellar autograft	24 (7–65) M	Contralateral knee

Continued

Table 1 Continued

Author, year	ACLR cohort	Graft type*	Time post-ACLR	Comparator cohort
Scanlan *et al*, 2013[61]	n=29, male/female 15/14, age 29±6 Y, height 1.7 ±0.1 m, weight 73±12 kg	n=29 Achilles tendon graft	27±4 M	Contralateral knee
Timoney *et al*, 1993[31]	n=10, male/female 10/0	n=10 patellar tendon graft	10 (8–12) Y	n=10, male/female 10/0
Wang *et al*, 2012[42]	Group 1 (Dominant limb ACLR∝): n=19, male/female 12/7, age 32±9 Y, height 1.8±0.1 m, weight 84±19 kg Group 2 (Non-dominant limb ACLR∝): n=22, male/female 12/10, age31±8 Y, height 1.7±0.1 m, weight 81±16 kg	Group 1: n=12 hamstring tendon graft, 7 patellar tendon graft Group 2: n=19 hamstring tendon graft, 3 patellar tendon graft	Group 1: 14±4 M Group 2: 14±6 M	n=20, male/female 13/7, age 23±3 Y height 1.8±0.1 m, weight 71±13 kg
Wang *et al*, 2013[38]	Group 1 (ACLR technique AMP): n=12, male/female 7/5, age 30±7 Y, height 1.7±0.1 m, weight 86±17 kg Group 2 (ACLR technique TT): n=12, male/female 7/5, age 33±7 Y, height 1.8±0.1 m, weight 83±17 kg	Group 1: n=12 hamstring tendon graft; Group 2: n=12 hamstring tendon graft	Group 1: 9±4 M Group 2: 17±11 M	n=20, male/female 15/5, age 26±8 Y, height 1.8±0.1 m, weight 82±16 kg
Webster *et al*, 2005[18]	Group 1 (Hamstring tendon graft): n=17, male/female 16/1, age 27±8 Y, height 1.8±0.1 m, weight 80±7 kg Group 2 (Patellar tendon graft): n=17, male/female 16/1, age 24±5 Y, height 1.8±0.1 m, weight 79 ±6 kg	Group 1: n=17 hamstring tendon graft Group 2: n=17 patellar tendon graft	Group 1: 9±2 MGroup 2: 11 ±2 M	n=17, male/female 16/1, age 25±5 Y, height 1.8±0.1 m, weight 75±12 kg
Webster *et al*, 2011[36]	Group 1 (Hamstring tendon graft): n=18, male/female 16/2, age 27±6 Y, height 1.8±0.1 m, weight 79±7 kgGroup 2 (Patellar tendon graft): n=18, male/female 17/1, age 24±6 Y, height 1.8±0.1 m, weight 79±7 kg	Group 1: n=18 hamstring tendon graft Group 2: n=18 patellar tendon graft	Group1: 9±2 M Group 2: 11±2 M	n=18, male/female 16/2, age 25±5 Y, height 1.8±0.12 m, weight 75±13 kg
Webster *et al*, 2012a[37]	Group 1 (Hamstring tendon graft): n=16, male/female 16/0, age 28±6 Y, height 1.8±0.1 m, weight 80±7 kgGroup 2 Patellar tendon graft): n=16, male/female 16/0, age 24±6 Y, height 1.8±0.1 m, weight 79±6 kg	Group 1: n=16 hamstring tendon graft Group 2: n=16 patellar tendon graft	Group1: 9±3 M Group 2: 11±2 M	n=16, male/female 16/0, age 25±5 Y, height 1.8±0.1 m, weight 76±12 kg
Webster *et al*, 2012[27]	Group 1 (Female): n=18, male/female 0/18, age 28 ±7 Y, height 1.7±0.1 m, weight 71±8 kg Group 2 (Male): n=18, male/female 18/0, age 27±7 Y, height 1.8±0.1 m, weight 81±13 kg	Group 1: n=18 hamstring tendon graft Group 2: n=18 hamstring tendon graft	Group1: 20±12 M Group 2: 20±12 M	Contralateral knee
Webster *et al*, 2012[25]	n=16, male/female 13/3, age 26±6 Y, height 1.8 ±0.1 m, weight 82±11 kg	n=8 hamstring tendon graft, 8 patellar tendon graft	10±2 M, 3±1 Y	Contralateral knee
Zabala *et al*, 2013[32]	n=45, male/female 26/19, age 30±6 Y, height 1.7 ±0.1 m, weight 74±12 kg	n=45 Achilles tendon autograft	26 (22–36) M	n=45, male/female 26/19, age 30±5 Y, height 1.7±0.1 m, weight 74±12 kg

Values are presented as mean±SD, or median (IQR). Male and female values are presented as numbers.
*Description of graft type taken directly from each study as described in text.
∝ Lower limb dominance defined by ball kicking; ACLR, anterior cruciate ligament reconstruction; AMP, anteromedial portal technique; BMI, body mass index; D, days; kg, kilograms; m, metres; M, months; n, number; RTS, return to sport; TT, transtibial technique; W, weeks; Y, years.

conducted based on patellar and hamstring tendon autograft types.

Based on the methodological quality ratings, a level of evidence was assigned to each variable of interest as follows as per van Tulder *et al*,[14]: (1) strong evidence provided by pooled results derived from three or more studies, including a minimum of two high-quality studies that were statistically homogenous (I^2 not significant at 0.05); may be associated with a statistically significant or non-significant pooled result; (2) moderate evidence provided by statistically significant pooled results derived from multiple studies that were statistically heterogeneous, including at least one high-quality study; or from multiple moderate-quality or low-quality studies which were statistically homogenous; (3) limited evidence provided by results from one high-quality study or multiple moderate-quality or low-quality studies that are statistically heterogeneous; (4) very limited evidence provided by results from one moderate-quality or low-quality study; and (5) no evidence provided by pooled results that are insignificant and derived from multiple statistically heterogeneous studies (regardless of quality).

RESULTS

Search strategy, methodological quality and risk of bias

The comprehensive search strategy identified 14 613 titles, with the last search conducted on 19 February 2014 (figure 1). Following removal of duplicate publications and conference proceedings, titles of 5114 publications were evaluated and 2959 publications were evaluated beyond title (ie, Abstract). The full texts of 52 articles were retrieved, with 34 articles meeting the selection criteria. Table 1 presents characteristics of the included studies. The methodological quality scores ranged from 3 to 13 (of 15), with an average score of 10 (see online supplementary table C). There were 6 studies of high quality, 18 were moderate quality and 10 were low quality. All 34 studies scored positively on item 5 (comparison group) and negatively on item 13 (adjustment for confounders). Only two studies reported sample size calculations.

Findings

Sagittal plane kinematics: comparison to healthy controls

In individuals 6 months post-ACLR, pooled data from two low-quality studies showed moderate evidence of greater peak knee flexion angle compared to healthy controls (SMD: 1.06, 95% CI 0.39 to 1.74; I^2=0%, p=0.54)[15,16] (table 2 and figure 2). At 6–12 months post-ACLR, pooled data from one moderate-quality study and one high-quality study showed moderate evidence of no difference in peak knee flexion angle between ACLR and control participants (−0.35, −0.82 to 0.11; I^2=0%, p=0.44).[17,18] Very limited evidence of smaller peak knee flexion angles in individuals with ACLR was provided by one moderate-quality study at 1–3 years postsurgery (−2.21, −3.16 to −1.26),[19] and one low-quality study at ≥3 years postsurgery, (−1.38, −2.14 to −0.62).[20] Two studies reported mean knee flexion angle in individuals who were 6 months post-ACLR,[21,22] and a single study reported peak angle at initial impact in individuals ≥3 years post-ACLR.[23] These studies were not included in meta-analyses.

Sagittal plane kinematics: comparison to contralateral knee

At 6–12 months after ACLR, pooled data from three studies of moderate and low quality provided limited evidence of smaller peak knee flexion angle in the ACLR knee compared to the contralateral knee (−1.74, −3.19 to −0.29; I^2=93%, p<0.0001)[24–26] (figure 3). At 1–3 years post-ACLR, there was moderate evidence from three studies (one high-quality, two low-quality) of no difference in peak knee flexion angle between the ACLR and contralateral limbs (−0.14, −0.44 to 0.15; I^2=0%, p=0.80).[26–28] At ≥3 years post-ACLR, a single moderate-quality study provided very limited evidence of no between-limb difference in peak knee flexion angle (0.02, −0.68 to 0.71)[25] (table 2). A single study reported mean knee flexion angles in individuals ≥3 years post-ACLR,[29] and thus it was not included.

Sagittal plane moments: comparison to healthy controls

Six months after ACLR, pooled data from two low-quality studies showed limited evidence of greater peak knee flexion moment (1.61, 0.87 to 2.35; I^2=0%, p=0.43), and lower knee extension moment (−3.55, −4.63 to −2.48; I^2=0%, p=0.33) than healthy controls[15,16] (table 3 and figure 2). In individuals who were 6–12 months post-ACLR, there was moderate evidence from pooled data from three studies (two low- and one high-quality) for smaller peak knee flexion moment than controls (−0.76, −1.40 to −0.12; I^2=52%, p=0.12),[18,30,31] and

Table 2 Sagittal plane knee kinematics

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95% CI)
ACLR vs healthy controls					
Peak knee flexion angle					
<6M	Ferber *et al*, 2002[15]	19.8±5.4°	3 M	16.1±2.1°	0.86 (−0.06 to 1.79)
	Ferber *et al*, 2003[16]	16.8±2.2°	3 M	13.7±2.4°	1.29 (0.31 to 2.27)
	Devita *et al*, 1998[59]	24°	3 W	22.2°	ACLR limb walked with more flexed knee (p value not reported)
6 to <12M	Georgoulis *et al*, 2003[17]	13.4±10.0°	30±17 W	14.5±5.8°	−0.12 (−0.87 to 0.63)
	Webster *et al*, 2005[18]	Hamstring-tendon graft: 21±4.9°	9±2 M	21.5±5.6°	−0.09 (−0.77 to 0.58)
		Patellar-tendon graft: 17.2±3.6°	11±2 M	21.5±5.6°	−0.89 (−1.60 to −0.18)
	Devita *et al*, *et al*, 1998[59]	20.5°	6 M	22.2°	No significant differences reported (p value not reported)
	Wang *et al*, 2013[38]	AMP technique: 20°	9±4 M	16°	No significant differences reported (p value not reported)
1 to <3Y	Gao and Zheng 2010[19]	16±2°	12 M	20±1.5°	−2.21 (−3.16 to −1.26)
	Bulgheroni *et al*, 1997[58]	16°		14°	Sagittal plane kinematics were maintained (p value not reported)
	Wang *et al*, 2013[38]	TT technique: 18°	17±11 M	16°	No differences reported
≥3Y	Patterson *et al*, 2014[40]	16.5°±2	4±3 Y	19°±1.5	−1.38 (−2.14 to −0.62)
ACLR vs contralateral Knee					
Peak knee flexion angle					
<6M	Hunt *et al*, 2010[60]	12.7°	31±12 D	9.1°	No reported
6 to <12M	Di Stasi *et al*, 2013[24]	RTS-Fail: 22.6±1.4°	6 M	29.1±1.4°	−4.56 (−5.72 to −3.40)
		RTS-Pass: 22.4±1.5°	6 M	25.1±1.5°	−1.76 (−2.51 to −1.02)
	Webster *et al*, 2012[25]	19.8±5.1°	10±2 M	20.8±5°	−0.19 (−0.89 to 0.50)
	Roewer *et al*, 2011[26]	23.2±6.6°	6±1 M	28.1±6.7°	−0.73 (−1.29, −0.16)
	Bacchini *et al*, 2009[57]	4.61°	6 (5–7) M	6.15°	No differences reported
	Hooper *et al*, 2002[35]	14.2°	6±1 M	19.2°	No differences reported
1 to <3Y	Webster *et al*, 2012[27]	Males: 19.1±4.5°	20±12 M	19.6±5.4°	−0.10 (−0.75 to 0.56)
		Females: 16.8±5.8°	20±12 M	16.2±6.5°	0.10 (−0.56 to 0.75)
	Roewer *et al*, 2011[26]	24.4±7°	2 Y	25.4±6.9°	−0.14 (−0.69 to 0.40)
	Scanlan *et al*, 2010[28]	16.8±1.63°	24(7–65) M	17.4±1.86°	−0.34 (−0.89 to 0.21)
	Hooper *et al*, 2002[35]	14.6°	12±1 M	19.6°	No differences reported
	Wang *et al*, 2012[42]	∝Dominant limb ACLR: 14.3°	14±4 M	14.1°	No differences reported
		∝Non-dominant limb ACLR: 14.3°	14±6 M	14.1°	No differences reported
≥3Y	Webster *et al*, 2012[25]	20.2±5.4°	3±1 Y	20.1±5.6°	0.02 (−0.68 to 0.71)

Data are presented as mean±SD.
∝, Lower limb dominance defined by ball kicking; ACLR, anterior cruciate ligament reconstruction; AMP, anteromedial portal; M, month; RTS, return to sport; TT, transtibial; W, week; Y, year.

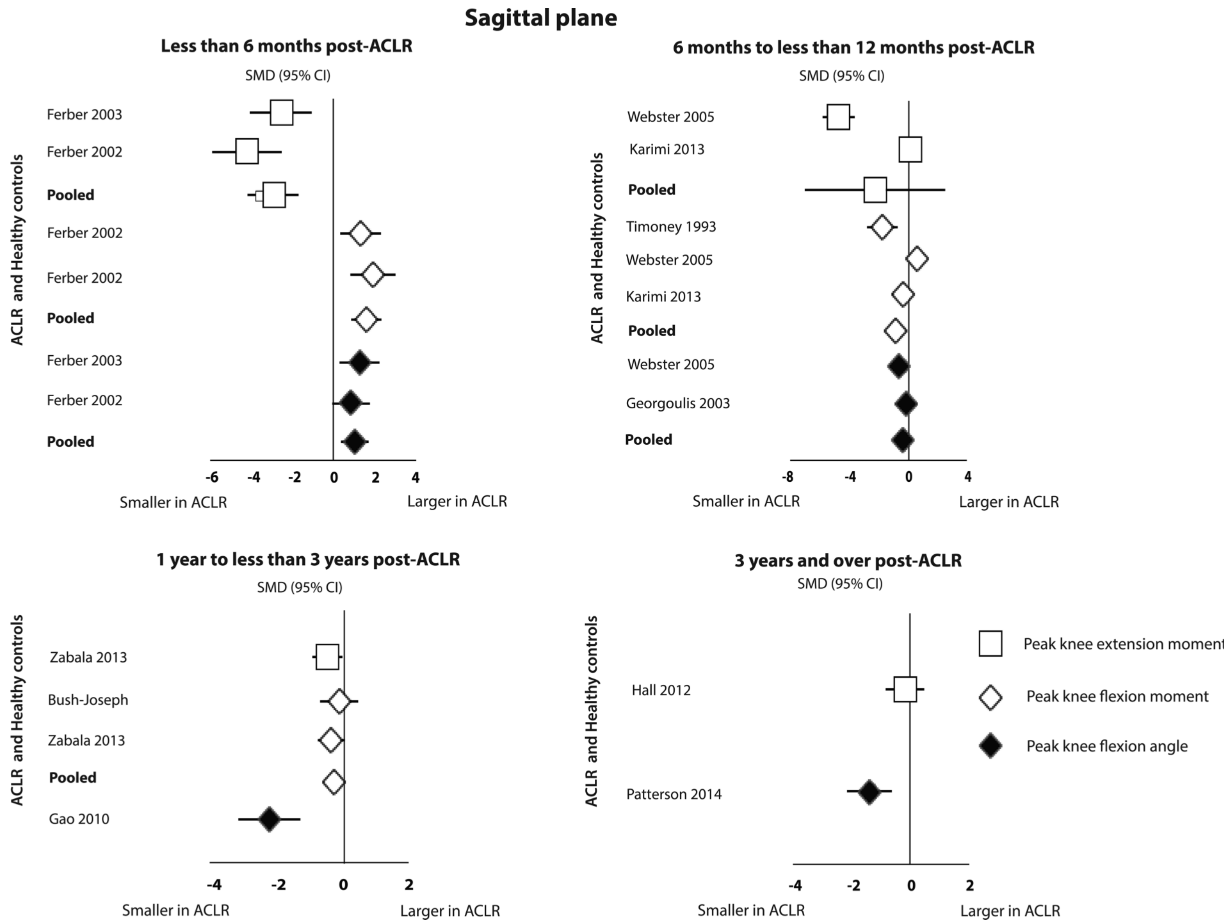

Figure 2 Standardised mean difference (SMD) for peak sagittal plane knee joint kinematics and moments. SMD >0: peak knee joint kinematics/moments larger in anterior cruciate ligament reconstruction (ACLR) participants compared to healthy controls. SMD <0: peak knee joint kinematics/moments smaller in ACLR participants compared to healthy controls.

limited evidence from two studies (one low- and one high-quality) for no difference in peak knee extension moment (−2.25, −7.00 to 2.50; I^2=98%, p<0.0001).[18 30] In individuals 1–3 years after ACLR, pooled data from two moderate-quality studies provided moderate evidence of no difference in peak knee flexion moment (−0.24, −0.58 to 0.10; I^2=0%, p=0.52),[32 33] while very limited evidence of no difference in peak knee extension moment (−0.40, −0.82 to 0.02) was provided by one moderate-quality study.[32] In those who were ≥3 years post-ACLR, a single high-quality study provided limited evidence of no difference in peak knee extension moment.[34] A single study in individuals who were ≥3 years post-ACLR reported sagittal plane moments at initial impact and this study was not included in the meta-analysis.

Sagittal plane moments: comparison to contralateral knee

In individuals who were 6–12 months after ACLR, we found limited evidence from pooled data of smaller knee flexion moments (−1.29, −2.45 to −0.12; I^2=93%, p<0.0001) from five studies (three moderate- and two low- quality),[24–26 30 35] and moderate evidence of smaller knee extension moments (−0.48, −0.93 to −0.03; I^2=0%, p=0.93) from three studies (two moderate- and one low-quality)[25 30 35] (figure 3). In individuals 1–3 years post-ACLR, moderate evidence of lower peak knee flexion moment (−0.27, −0.53 to −0.01; I^2=0%, p=0.98) was evident from pooling data from one high-quality, two moderate-quality, and one low-quality study,[26 27 32 35] and lower peak knee extension moment (−0.55, −0.93 to −0.16; I^2=0%, p=0.54) from pooling data from two moderate-quality studies.[32 35] In those who were ≥3 years post-ACLR, pooled data from two studies (one high- and one moderate-quality) provided moderate evidence of no difference in peak knee extension moment between ACLR and contralateral limbs (−0.11, 95% CI −0.59 to 0.36; I^2=0%, p=0.42),[25 34] while a single moderate-quality study provided very limited evidence of no between-limb difference in peak knee flexion moment (−0.20, −0.90 to 0.49)[25] (figure 3 and table 3).

Frontal plane kinematics: comparison to healthy controls

In those who were 6–12 months after ACLR, pooled data from four studies (two moderate- and two high-quality) provided moderate evidence of no difference in peak knee adduction (varus) angle in individuals with ACLR compared to controls (−0.43, −0.91 to 0.05; I^2=51%, p=0.10)[17 36–38] (table 4 and figure 4). In those who were 1–3 years post-ACLR, a single high-quality study provided limited evidence of lower peak knee adduction angle compared to controls (−1.10, −1.88 to −0.33).[38] In individuals who were ≥3 years post-ACLR, pooled data from one moderate- and one low-quality study provided limited evidence of no difference in peak adduction angle when compared to controls (1.08, −0.41 to 2.56; I^2=87%, p=0.005)[20 39] (figure 4).

Frontal plane kinematics: comparison to contralateral knee

Very limited evidence of no difference in peak knee adduction angle between individuals following ACLR and contralateral

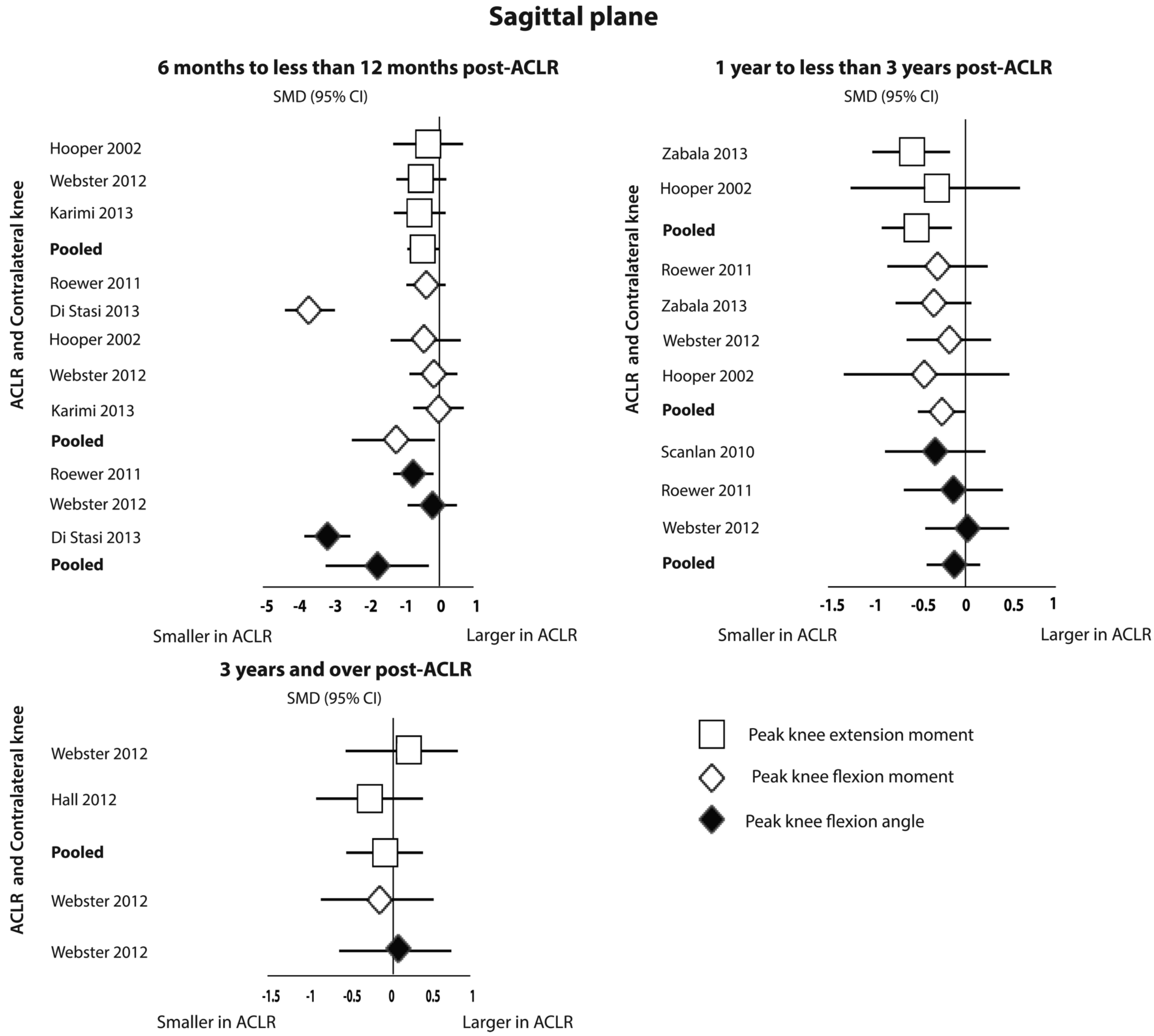

Figure 3 Standardised mean difference (SMD) for peak sagittal plane knee joint kinematics and moments. SMD >0: peak knee joint kinematics/moments larger in anterior cruciate ligament reconstruction (ACLR) knees compared to contralateral knees. SMD <0: peak knee joint kinematics/moments smaller in ACLR knees compared to contralateral knees.

knees (0.22, −0.47 to 0.92) was provided from a single study of moderate-quality in those who were 6–12 months after ACLR,[25] and limited evidence of no difference in those who were 1–3 years post-ACLR, from a single high-quality study (0.02, −0.44 to 0.48)[27] (figure 5 and table 4). In individuals who were ≥3 years post-ACLR, a moderate-quality study provided very limited evidence of no difference in peak knee adduction angle between ACLR and contralateral knees (0.02, −0.67 to 0.72)[25] (figure 5).

Frontal plane moments: comparison to healthy controls

In people who were 6–12 months after ACLR, pooled data from two studies (one low-quality, one high-quality) provided moderate evidence of no difference in peak knee adduction moment (−0.61, −2.71 to 1.49; I^2=94%, p<0.0001)[30 37] (table 5 and figure 4). One to 3 years post-ACLR, data from a single moderate-quality study provided very limited evidence of no difference in peak knee adduction moment (−0.39, 95% −0.81 to 0.03).[32] In those who were ≥3 years post-ACLR, pooled data from four studies (two high-, one moderate- and one low-quality) provided strong evidence of no difference in peak knee adduction moment when compared to control participants (0.09, −0.63 to 0.81; I^2=75%, p=0.008)[34 39–41] (figure 4).

Frontal plane moments: comparison to contralateral knee

From meta-analyses, we found moderate evidence of no difference in peak knee adduction moment between ACLR and contralateral knees at multiple time points after ACLR (figure 5 and table 5). Pooled data from four studies (one low-, two moderate- and one high-quality) found no between-limb difference in those who were 6–12 months after ACLR (−0.27, −0.60 to 0.06; I^2=0%, p=0.74);[25 30 35 37] four studies (three moderate- and one high- quality) in people 1–3 years post-ACLR (−0.18, −0.53 to 0.16; I^2=21%, p=0.29);[27 32 35 42] and three studies (one low-, one moderate- and one high-quality) ≥3 years post-ACLR (−0.09, −0.51 to 0.32; I^2=0%, p=0.55)[25 34 41] (figure 5).

Transverse plane kinematics: comparison to healthy controls

Due to the large heterogeneity in reporting of transverse plane kinematics data, meta-analyses could not be performed (table 6

Table 3 Sagittal plane knee moments

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95% CI
ACLR vs healthy controls					
Peak knee flexion moment					
<6M	Devita *et al*, 1997[22]	25.8 Nm	3 W	45.2 Nm	Lower flexion moment in ACLR knee throughout stance
		29.0 Nm	5 W	45.2 Nm	Not reported
	Devita *et al*, 1998[59]	25.3 Nm	3 W	63.1 Nm	Not reported
	Ferber *et al*, 2002[15]	0.35±0.06 Nm/kg	3 M	0.17±0.11 Nm/kg	1.95 (0.84 to 3.05)
	Ferber *et al*, 2003[16]	0.38±0.06 Nm/kg	3 M	0.22±0.15 Nm/kg	1.34 (0.35 to 2.33)
6 to <12M	Devita *et al*, 1998[59]	37.9 Nm	6 M	63.1 Nm	Smaller flexion moment in ACLR knee
	Karimi *et al*, 2013[30]	0.45±0.35 Nm/kg	6 M	0.61±0.5 Nm/kg	−0.36 (−1.08 to 0.36)
	Webster *et al*, 2005[18]	Hamstring graft: 2.7±1.3 Nm/(%kg m)	9±2 M	2.8±1.0 Nm/(%kg m)	−0.08 (−0.76 to 0.59)
		Patellar graft: 1.7±0.7 Nm/(%kg m)	11±2 M	2.8±1.0 Nm/(%kg m)	−1.24 (−1.99 to −0.50)
	Timoney *et al*, 1993[31]	2.02±1.14 Nm/(%kg m)	10 (8–13) M	3.74±0.82 Nm/(%kg m)	−1.66 (−2.71 to −0.61)
1 to <3Y	Zabala *et al*, 2013[32]	2.2±1.3 Nm/(%kg m)	26 (22–36) M	2.6±1.2 Nm/(%kg m)	−0.32 (−0.73 to 0.10)
	Bush-Joseph *et al*, 2001[33]	2.4±1.0 Nm/(%kg m)	22±12 M	2.5±1.5 Nm/(%kg m)	−0.08 (−0.67 to 0.51)
	Bulgheroni *et al*, 1997[58]	0.55 Nm/(%kg m)	17±5 M	0.42 Nm/(%kg m)	No difference during the stance; however increased moment upon landing in ACLR knee
≥3Y	Noehren *et al*, 2013[23]	0.12 Nm/kg	5±3 Y	0.32 Nm/kg	Not reported. Flexion moment at initial landing significantly lower in ACLR knee
	Ferber *et al*, 2004[29]	0.35 Nm/kg	6±5 Y	0.19 Nm/kg	Not reported
ACLR vs healthy controls					
Peak extension moment					
<6M	Devita *et al*, 1997[22]	−2.5 Nm	3 W	25 Nm	Not reported
		0 Nm	5 W	25 Nm	Not reported
	Devita *et al*, 1998[59]	−5 Nm	3 W	17.5 Nm	Not reported
	Ferber 2002[15]	0.11±0.08 Nm/kg	3 M	0.44±0.07 Nm/kg	−4.20 (−5.90 to −2.51)
	Ferber *et al*, 2003[16]	−0.03±0.01 Nm/kg	3 M	0.27±0.13 Nm/kg	−3.12 (−4.51 to −1.73)
6 to <12M	Devita *et al*, 1998[59]	17.5 Nm	6 M	17.5 Nm	Not reported
	Karimi *et al*, 2013[30]	0.55±0.44 Nm/kg	6 M	0.50±0.10 Nm/kg	0.15 (−0.56 to 0.87)
	Webster *et al*, 2005[18]	Hamstring graft: 2.0±0.07 Nm/(%kg m)	9±2 M	2.6±0.07 Nm/(%kg m)	−8.37 (−10.59 to −6.15)
		Patellar graft: 2.5±0.08 Nm/(%kg m)	11±2 M	2.6±0.07 Nm/(%kg m)	−1.30 (−2.05 to −0.55)
1 to <3Y	Zabala *et al*, 2013[32]	2.8±0.96 Nm/(%kg m)	26 (22–36) M	3.17±0.88 Nm/(%kg m)	−0.40 (−0.82 to 0.02)
≥3Y	Hall *et al*, 2012[34]	0.42±0.21 Nm/kg	6 (2–18) Y	0.45±0.14 Nm/kg	−0.16 (−0.82 to 0.49)
	Ferber *et al*, 2004[29]	0.13 Nm/kg	6±5 Y	0.45 Nm/kg	No reported
	Noehren 2013[23]	0.40 Nm/kg m	5±3 Y	0.22 Nm/kg	No reported
ACLR vs healthy controls					
Peak flexion moment					
6 to <12M	Bacchini *et al*, 2009[57]	32.2 Nm	6 (5–7) M	23.3 Nm	Described higher flexion in ACLR knee (p value not reported)
	Karimi *et al*, 2013[30]	0.45±0.35 Nm/kg	6 M	0.46±0.46 Nm/kg	−0.02 (−0.74 to 0.69)
	Webster *et al*, 2012[37]	0.39±0.24 Nm/kg	10±2 M	0.43±0.26 Nm/kg	−0.16 (−0.85 to 0.54)
	Hooper *et al*, 2002[35]	0.30±0.13 Nm/kg m	6±1 M	0.35±0.12 Nm/kg m	−0.38 (−1.37 to 0.61)
	Di Stasi *et al*, 2013[24]	RTS-Fail ACLR: 0.37±0.03 Nm/kg m	6 M	0.52±0.03 Nm/kg m	−4.91 (−6.14 to −3.68)
	Di Stasi *et al*, 2013[24]	RTS-Pass ACLR: 0.42±0.03 Nm/kg m	6 M	0.49±0.03 Nm/kg m	−2.29 (−3.10 to −1.47)
	Roewer *et al*, 2011[26]	0.40±0.12 Nm/BMI	6 M	0.49±0.32 Nm/BMI	−0.37 (−0.92 to 0.18)
1 to <3Y	Hooper *et al*, 2002[35]	0.31±0.16 Nm/kg m	12±1 M	0.38±0.12 Nm/kg m	−0.47 (−1.41 to 0.47)
	Webster *et al*, 2012[27]	Males: 0.27±0.12 Nm/kg m	20±12 M	0.29±0.13 Nm/kg m	−0.16 (−0.81 to 0.50)
		Females: 0.26±0.11 Nm/kg m	20±12 M	0.28±0.12 Nm/kg m	−0.17 (−0.82 to 0.48)
	Zabala *et al*, 2013[32]	2.23±1.29 Nm/(%kg m)	26 (22–36) M	2.59±1.17 Nm/(%kg m)	−0.31 (−0.73 to 0.10)
	Roewer 2011[26]	0.48±0.18 Nm/BMI	2 Y	0.53±0.16 Nm/BMI	−0.29 (−0.84 to 0.26)

Continued

Table 3 Continued

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95% CI
≥3Y	Webster *et al*, 2012[25]	0.41±0.23 Nm/kg	3±1 Y	0.46±0.25 Nm/kg	−0.20 (−0.90 to 0.49)
	Ferber *et al*, 2004[29]	0.35 Nm/kg	6±5 Y	0.60 Nm/kg	Not reported
Peak extension moment					
6 to <12M	Bacchini *et al*, 2009[57]	3.2 Nm	6 (5–7) M	1.72 Nm	Described higher extension moment in ACLR knee (p vale not reported)
	Karimi *et al*, 2013[30]	0.55±0.44 Nm/kg	6 M	0.88±0.70 Nm/kg	−0.55 (−1.28 to 0.18)
	Webster *et al*, 2012[25]	0.40±0.13 Nm/kg	10±2 M	0.47±0.14 Nm/kg	−0.51 (−1.21 to 0.20)
	Hooper *et al*, 2002[35]	0.09±0.14 Nm/kg m	6±1 M	0.14±0.16 Nm/kg m	−0.31 (−1.30 to 0.67)
1 to <3Y	Hooper *et al*, 2002[35]	0.16±0.11 Nm/kg m	12±1 M	0.19±0.09 Nm/kg m	−0.28 (−1.21 to 0.65)
	Zabala 2013[32]	2.8±0.96 Nm/(%kg m)	26 (22–36) M	3.36±0.88 Nm/(%kg m)	−0.60 (−1.03 to −0.18)
≥3Y	Hall *et al*, 2012[34]	0.42±0.21 Nm/kg	6 (2–18) Y	0.50±0.31 Nm/kg	−0.30 (−0.96 to 0.36)
	Webster *et al*, 2012[25]	0.51±0.11 Nm/kg	3±1 Y	0.50±0.10 Nm/kg	0.09 (−0.60 to 0.79)
	Ferber *et al*, 2004[29]	0.13 Nm/kg	6±5 Y	0.25 Nm/kg	Not reported

Data are presented as mean±SD.
ACLR, anterior cruciate ligament reconstruction; AMP, anteromedial portal; kg, kilograms; m, metres; M, month; Nm, Newton metres; RTS, return to sport; TT, transtibial; W, week; Y, year.

and figure 6). In those who were 6–12 months post-ACLR, one moderate-quality study provided very limited evidence of lower peak internal rotation angle at midstance (27–67% of stance) compared to controls (−1.01, −1.61 to −0.41).[36] One high-quality study provided limited evidence of higher peak knee internal rotation angle before the toe-off phase of stance in the ACLR group (1.14, 0.36 to 1.91).[38] Two studies provide evidence of no difference in peak knee external rotation angle towards the end of stance phase between ACLR and control participants. One high-quality study provides limited evidence of no difference before toe-off (−0.37, −1.09 to 0.35),[38] while a moderate-quality study provides very limited evidence of no difference during toe-off (−0.36, −1.12 to 0.40).[17] In those who were 1–3 years after ACLR, one high-quality study provided limited evidence of increased peak knee internal rotation angle before toe off (1.18, 0.40 to 1.96) and no difference in peak knee external rotation angle before toe-off, compared to controls (−0.73, −1.47 to 0.01).[38] In participants ≥3 years post-ACLR, a single study of low-quality provided very limited evidence of smaller peak knee internal rotation angle

Table 4 Frontal plane knee kinematics

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95% CI)
ACLR vs healthy controls					
Peak knee adduction angle					
6 to <12M	Georgoulis *et al*, 2003[17]	1.7±4.1°	30±17 W	1.0±3.1°	0.18 (−0.58 to 0.93)
	Wang *et al*, 2013[38]	AMP technique: −2.5±2°	9±4 M	−2.3±1.5°	−0.11 (−0.83 to 0.60)
	Webster *et al*, 2011[36]	Hamstring graft: 3.7±3.3°	9±2 M	7.7±2.8°	−1.28 (−2.00 to −0.55)
		Patellar graft: 6.5±3.0°	11±2 M	7.7±2.8°	−0.40 (−1.07 to 0.26)
	Webster *et al*, 2012[37]	Hamstring graft: 3.9±3.7°	9±2 M	7.7±2.8°	−1.13 (−1.88 to −0.38)
		Patellar graft: 6.6±2.8°	11±2 M	7.7±2.8°	−0.38 (−1.08 to 0.32)
1 to <3Y	Wang *et al*, 2013[38]	AMP technique: −4±1.5°	9±4 M	−2.3±1.5°	−1.10 (−1.88 to −0.33)
	Bulgheroni *et al*, 1997[58]	5°	17±5 M	3.8°	No differences reported
	Gao and Zheng 2010[19]	1.0°	12 M	−3.5°	No differences reported
≥3Y	Butler *et al*, 2009[39]	1.9±2.9°	5±3 Y	0.9±2.9°	0.34 (−0.34 to 1.01)
	Patterson *et al*, 2014[20]	0.5±1°	4±3 Y	−1.0±0.5°	1.85 (1.03 to 2.67)
ACLR vs contralateral knee					
Peak knee adduction angle					
6 to <12M	Webster *et al*, 2012[25]	7.2±4.9°	10±2 M	6.1±4.7°	0.22 (−0.47 to 0.92)
1 to <3Y	Webster *et al*, 2012[27]	Males: 5.4±2.9°	20±12 M	5.0±4.9°	0.10 (−0.56 to 0.75)
		Females: 4.8±3.1°	20±12 M	5.0±3.5°	−0.06 (−0.71 to 0.59)
	Wang *et al*, 2012[42]	∝Dominant limb ACLR: −1.5°	14±4 M	1.0°	No differences reported
		∝Non-dominant limb ACLR : −2.0°	14±6 M	−3.5°	No differences reported
≥3Y	Webster *et al*, 2012[25]	8.2±4.1°	3±1 Y	8.1±4.3°	0.02 (−0.67 to 0.72)

Data are presented as mean±SD.
∝, Lower limb dominance defined by ball kicking, ACLR, anterior cruciate ligament reconstruction; AMP, anteromedial portal; M, month; RTS, return to sport; TT, transtibial; W, week; Y, year.

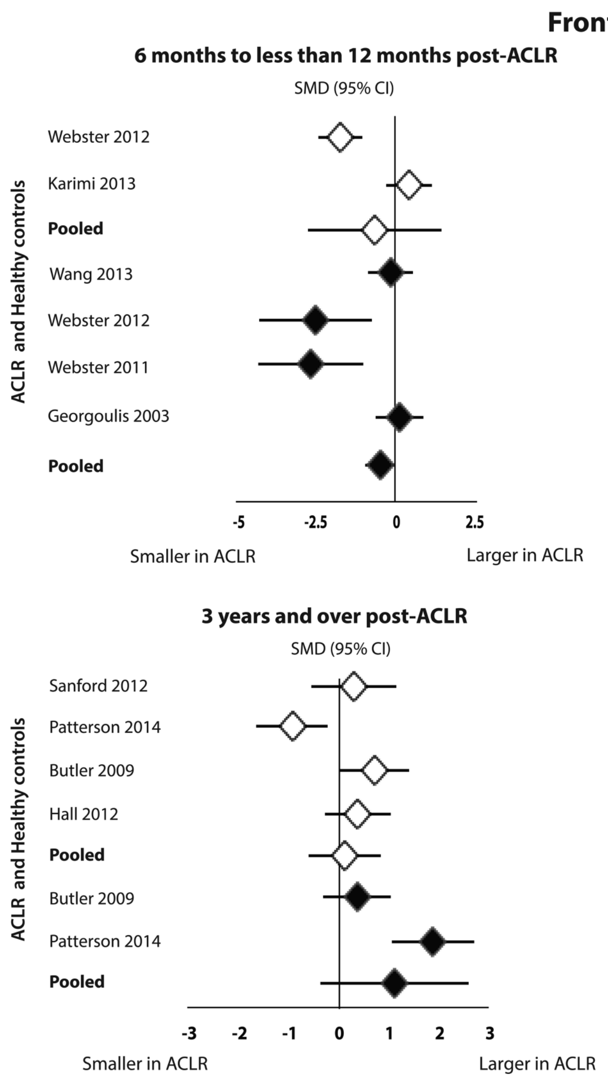

Figure 4 Standardised mean difference (SMD) for peak frontal plane knee joint kinematics and moments. SMD >0: peak knee joint kinematics/moments larger in anterior cruciate ligament reconstruction (ACLR) participants compared to healthy controls. SMD <0: peak knee joint kinematics/moments smaller in ACLR participants compared to healthy controls.

before toe-off compared to controls (−1.66, −2.45 to −0.86)[20] (figure 6).

Transverse plane kinematics: comparison to contralateral knee

Very limited evidence was found for transverse plane kinematics in the ACLR knee compared to the contralateral knee (figure 7 and table 6). In people 6–12 months post-ACLR, a single study of moderate-quality reported no significant between-limb difference in peak knee internal rotation angle at midstance (−0.68, −1.40 to 0.03).[25] No difference in peak knee internal rotation angle between ACLR and contralateral knees was reported by a moderate-quality study in people who were 1–3 years after ACLR (−1.03, −2.16 to 0.11),[43] and ≥3 years post-ACLR (−0.46, −1.16 to 0.24)[25] (figure 7).

Transverse plane moments: comparison to healthy controls

In individuals who were 6–12 months post-ACLR, data from a single study of low-quality provided very limited evidence of no differences in peak knee external rotation moment (20% of stance) (0.00, −0.72 to 0.72) and peak knee internal rotation moment (80% of stance) (0.39, −0.33 to 1.12) between ACLR and control participants[30] (table 7 and figure 6). In those who were 1–3 years after ACLR, a single moderate-quality study provided very limited evidence of significantly lower peak knee external rotation moment at 25% of stance (−0.66, −1.09 to −0.24), but no difference in peak knee internal rotation moment at 75% of stance phase (−0.17, −0.58 to 0.25)[32] (figure 6).

Transverse plane moments: comparison to contralateral knee

In individuals who were 6–12 months after ACLR, a single low-quality study provided very limited evidence of significantly greater peak knee internal rotation moment in ACLR knees than contralateral knees (1.06, 0.29 to 1.83), but no difference in peak knee external rotation moment (20% of stance) (−0.07, −0.17 to 0.03)[30] (table 7 and figure 7). In those who were 1–3 years post-ACLR, we found very limited evidence from a single moderate-quality study of no difference in peak knee internal rotation moment at 75% of stance (−0.31, −0.72 to 0.11) or peak knee external rotation moment at 25% of stance (−0.26, −0.67 to 0.16) between ACLR and contralateral knees[32] (figure 7).

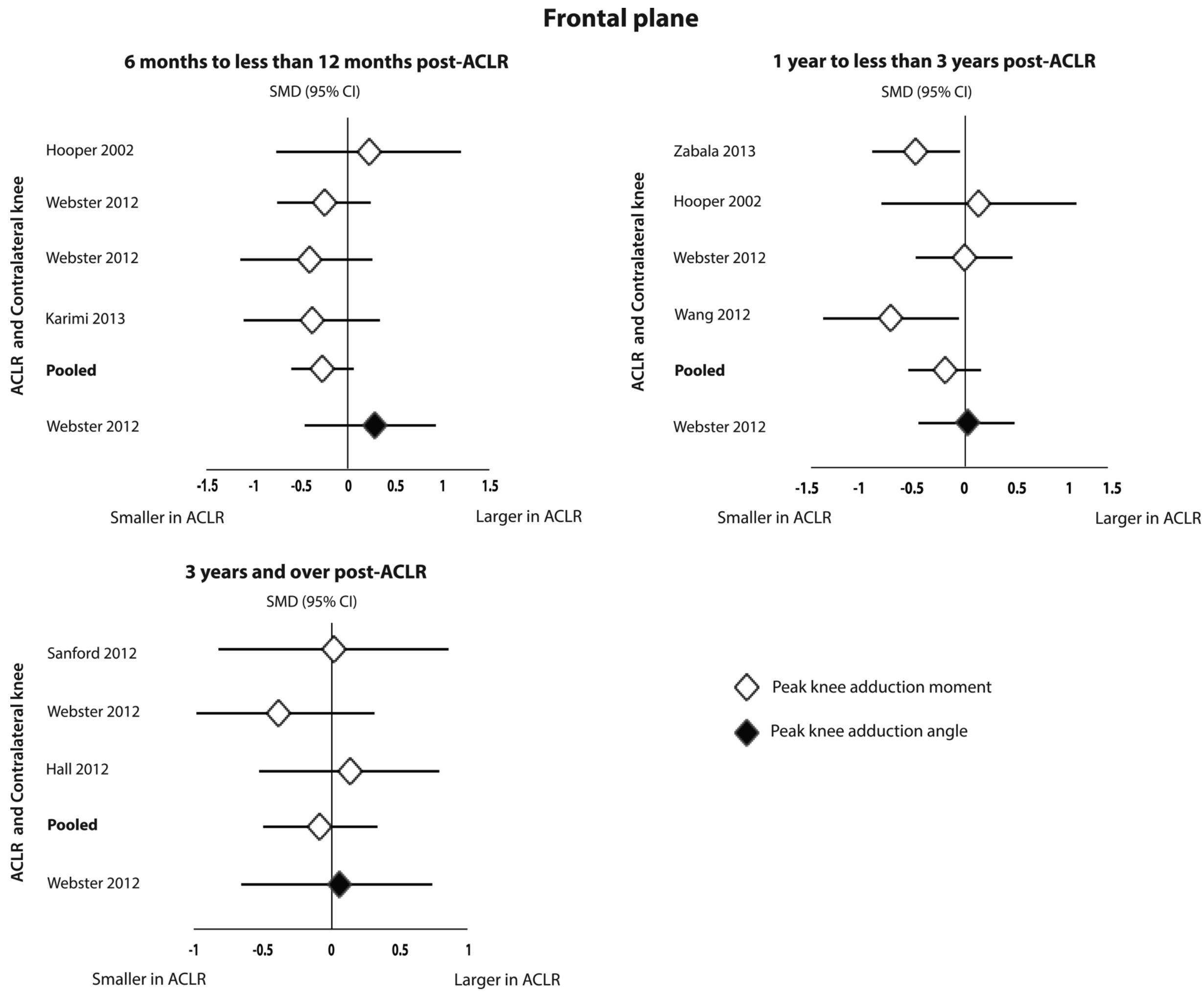

Figure 5 Standardised mean difference (SMD) for peak frontal plane knee joint kinematics and moments. SMD >0: peak knee joint kinematics/moments larger in anterior cruciate ligament reconstruction (ACLR) knees compared to contralateral knees. SMD <0: peak knee joint kinematics/moments smaller in ACLR knees compared to contralateral knees.

Sensitivity analysis based on graft type

We observed some differences in outcomes when sensitivity analyses based on hamstring- and patellar-tendon graft types were conducted in those 6–12 months after ACLR compared to healthy controls. The difference in peak knee flexion moment remained significant when patellar-tendon graft data were evaluated separately (−1.38, −1.99 to −0.78).[18 31] In contrast, the difference in peak knee flexion moment between ACLR and control participants was not significant when hamstring-tendon graft data were evaluated separately (−0.08, −0.76 to 0.59).[18] Compared to the combined pooled data, peak knee extension moment was significantly lower in ACLR participants with patellar-tendon grafts (−1.30, −2.05 to −0.55) and hamstring-tendon grafts (−8.37, −10.59 to −6.15) than control participants.[18] We also observed a smaller peak adduction angle in ACLR participants with hamstring-tendon grafts compared to controls (−1.07, −1.50 to −0.64);[36–38] however, no differences were observed when comparing ACLR participants with patellar-tendon grafts to control participants (−0.23, −2.05 to −0.55).[17 36 37] There was no effect of graft type on observed comparisons between ACLR and contralateral limbs at 6–12 months postsurgery. Sensitivity analyses revealed no effects of graft type on outcomes at 1–3 years or ≥3 years after ACLR, for comparisons between ACLR and control participants or between ACLR and uninjured contralateral limbs.

DISCUSSION

Knee kinematics and moments associated with ACLR were evaluated systematically, and the 34 studies included for data synthesis revealed that ACLR does not restore normal knee joint gait biomechanics. The results of this systematic review indicate that biomechanical deficits are evident, mostly in the sagittal plane from 6 months post-ACLR surgery.

The most consistent finding from this systematic review was of altered sagittal plane kinematics and moments. Gait characteristics observed during the early post-ACLR period (<6 months) do not appear to match those seen in individuals who are at least 6 months postsurgery. We speculate that greater knee flexion angles and, in turn, higher flexion moment observed throughout the gait cycle in individuals <6 months post-ACLR may reflect adapted walking pattern as a result of joint swelling, pain or muscle inhibition due to knee pain. Patellofemoral joint force is primarily influenced by flexion angles and moments.[44] Thus, increased knee flexion angles and moments during the early post-ACLR period is of clinical concern, since this may initiate degenerative processes in the joint. Our recent research identifies early PFJ OA changes at

Table 5 Frontal plane knee moments

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95%CI)
ACLR vs healthy controls					
Peak adduction moment					
6 to <12M	Karimi *et al*, 2013[30]	0.42±0.16 Nm/kg	6 M	0.35±0.13 Nm/kg	0.47 (−0.26 to 1.19)
	Webster *et al*, 2012[37]	Hamstring graft 0.27±0.08 Nm/kg m	9±2 M	0.40±0.08 Nm/kg m	−1.59 (−2.35 to −0.83)
		Patellar graft 0.28±0.06 Nm/kg m	11±2 M	0.40±0.08 Nm/kg m	−1.66 (−2.43 to −0.89)
1 to <3Y	Zabala *et al*, 2013[32]	2.67±0.64 Nm/(%kg m)	26(22–36) M	2.94±0.73 Nm/(%kg m)	−0.39 (−0.81 to 0.03)
	Bulgheroni *et al*, 1997[58]	0.58 Nm/(%kg m)	17±5 M	0.55 Nm/(%kg m)	No differences (p value not reported)
≥3Y	Hall *et al*, 2012[34]	0.5±0.13 Nm/kg	6(2–18) Y	0.46±0.09 Nm/kg	0.35 (−0.31 to 1.01)
	Butler *et al*, 2009[39]	0.36±0.08 Nm/kg m	5±3 Y	0.30±0.09 Nm/kg m	0.69 (−0.01 to 1.38)
	Patterson *et al*, 2014[40]	0.31±0.08 Nm/kg m	4±3 Y	0.41±0.12 Nm/kg m	−0.96 (−1.67 to −0.24)
	Sanford *et al*, 2012[41]	0.24±0.09 Nm/kg m	7±6 Y	0.22±0.05 Nm/kg m	0.27 (−0.57 to 1.12)
ACLR vs contralateral knee					
Peak adduction moment					
6 to <12M	Bacchini *et al*, 2009[57]	24.6 Nm	6 (5–7) M	18.4 Nm	Higher knee adduction moment in ACLR knee (p value not reported)
	Karimi *et al*, 2013[30]	0.42±0.16 Nm/kg	6 M	0.51±0.27 Nm/kg	−0.39 (−1.12 to 0.33)
	Webster *et al*, 2012[25]	0.52±0.12 Nm/kg	10±2 M	0.58±0.14 Nm/kg	−0.45 (−1.15 to 0.25)
	Webster *et al*, 2012[37]	Hamstring graft 0.27±0.08 Nm/kg m	9±2 M	0.31±0.1 Nm/kg m	−0.43 (−1.09 to 0.23)
		Patellar graft 0.28±0.06 Nm/kg m	11±2 M	0.28±0.06 Nm/kg m	0.00 (−0.65 to 0.65)
	Hooper *et al*, 2002[35]	0.16±0.04 Nm/kg m	6±1 M	0.15±0.05 Nm/kg m	0.21 (−0.77 to 1.19)
1 to <3Y	Wang *et al*, 2012[42]	∝Dominant limb ACLR 0.35 Nm/kg±0.16	14±4 M	0.46 Nm/kg±0.15	−0.70 (−1.34 to −0.05)
		∝Non-dominant limb ACLR 0.33 Nm/kg	14±6 M	0.45 Nm/kg	
	Webster *et al*, 2012[27]	Males 0.31±0.07 Nm/kg m	20±12 M	0.33±0.08 Nm/kg m	−0.26 (−0.92 to 0.40)
		Females 0.38±0.08 Nm/kg m	20±12 M	0.36±0.10 Nm/kg m	0.22 (−0.44 to 0.87)
	Hooper *et al*, 2002[35]	0.19±0.04 Nm/kg m	12±1 M	0.18±0.05 Nm/kg m	0.21 (−0.72 to 1.14)
	Zabala *et al*, 2013[32]	2.67±0.64 Nm/(%kg m)	26 (22–36) M	2.95±0.57 Nm/(%kg m)	−0.46 (−0.88 to −0.04)
≥3Y	Hall *et al*, 2012[34]	0.5±0.13 Nm/kg	6 (2–18) Y	0.48±0.2 Nm/kg	0.12 (−0.54 to 0.77)
	Webster *et al*, 2012[25]	0.56±0.15 Nm/kg	3±1 Y	0.63±0.19 Nm/kg	−0.40 (−1.10 to 0.30)
	Sanford *et al*, 2012[41]	0.24±0.09 Nm/kg m	7±6 Y	0.24±0.06 Nm/kg m	0.00 (−0.84 to 0.84)

Data are presented as mean±SD.
∝, Limb dominance defined by ball kicking; ACLR, anterior cruciate ligament reconstruction; M, month; Nm, Newton metres; Y, year.

1 year after ACLR,[45 46] which may be partly attributable to increased knee flexion moments in the first year following ACLR. Thus, targeting altered sagittal plane knee joint angles and moments during early rehabilitation may slow the development and progression of post-traumatic knee OA. However, longitudinal data are required in this patient population before clinical implications can be conclusively determined.

Following the early post-operative period (more than 6 months post-ACLR), individuals may walk with lower knee flexion angles and moments compared to healthy controls and contralateral knees. This finding is consistent with previous systematic reviews[5 6] that reported lower peak knee flexion angles and moments in individuals after ACLR compared to healthy controls. An external knee flexion moment causes the knee to flex and, in response, knee extensor muscles activate to produce an internal moment to resist the external flexion moment. It is plausible that lower knee flexion angles and moments evident in individuals >6 months post-ACLR may be related to altered quadriceps and/or hamstring muscle activation pattern in this patient population. However, this is speculative, since this systematic review did not investigate muscle activation patterns.

In the current systematic review, the most consistent findings were observed in individuals who were 6–12 months post-ACLR. However, this may also reflect the time point chosen for most studies. Very limited evidence (ie, single studies) indicates that lower peak knee flexion angles and moments may also be present in individuals 12 months or more post-ACLR. Sagittal plane gait characteristics may have important longer-term implications in this patient population. However, due to a paucity of studies beyond 12 months post-ACLR, we cannot draw any concrete conclusions. Longitudinal data from pre-injury to longer-term post-ACLR (~10 years) will assist in determining potentially modifiable risk factors associated with ACLR, which could be targeted during rehabilitation.

The knee adduction moment is an important risk factor for progression of non-traumatic knee OA,[47–49] and is targeted in many OA interventions.[50 51] However, this systematic review provides moderate evidence in individuals less than 3 years post-ACLR,[30 37 32] and strong evidence in those 3 years or longer post-ACLR,[34 39–41] that people who have undergone ACLR maintain similar knee adduction moments to healthy controls. Similarly, this systematic review also provided moderate evidence of no differences in peak knee adduction moments between ACLR and contralateral knees. Therefore, it is plausible that the knee adduction moment is less important in the development of knee OA associated with ACLR. This finding may reflect the compartment-specific prevalence of post-traumatic knee OA, with higher prevalence of lateral knee OA noted in post-traumatic knee OA than in non-traumatic

Table 6 Transverse plane knee kinematics

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95%CI)
ACLR vs healthy controls					
Peak internal rotation angle					
6 to <12M	Webster *et al*, 2011[36]	Hamstring graft: 7.0±5.0°	9±2 M	13.5±6.5°	−1.10 (−1.80 to −0.39)
		Patellar graft: 8.0±6.0°	11±2 M	13.5±6.5°	−0.86 (−1.55 to −0.17)
	Wang *et al*, 2013[38]	AMP technique: 4.7±0.9°	9±4 M	3±1.7°	1.14 (0.36 to 1.91)
1 to <3Y	Wang *et al*, 2013[38]	TT technique: 4.8±1.0°	17±11 M	3±1.7°	1.18 (0.40 to 1.96)
	Gao and Zheng 2010[19]	4.9°	12 M	2.5°	Described greater internal rotation in ACLR knee (p value not reported)
≥3Y	Patterson *et al*, 2014[20]	6.0±1.2°	4±3 Y	9.0±1.5	−1.66 (−2.45 to −0.86)
Peak external rotation angle					
6 to <12M	Georgoulis *et al*[17]	0.3±9.9°	30±17 W	3.6±6.2°	−0.36 (−1.12 to 0.40)
	Wang *et al*, 2013[38]	AMP technique: 5.0±2.3°	9±4 M	5.8±2°	−0.37 (−1.09 to 0.35)
1 to <3Y	Wang *et al*, 2013[38]	TT technique: 4.3±2.0°	17±11 M	5.8±2°	−0.73 (−1.47 to 0.01)
	Bulgheroni 1997[58]	6.7°	17±5 M	4.2°	No differences reported
	Gao and Zheng 2010[19]	6.0°	12 M	8.5°	No differences reported
ACLR vs contralateral knee					
Peak internal rotation angle					
6 to <12M	Webster *et al*, 2012[25]	9.5±5.7°	10±2 M	13.1±4.5°	−0.68 (−1.40 to 0.03)
1 to <3Y	Sato 2013[43]	3.3±4.4°	14±2 M	8.4±4.9°	−1.03 (−2.16 to 0.11)
	Scanlan *et al*, 2010[28]	3.5°	24 (7–65) M	5.5°	Described lower internal rotation in ACLR knee (p value not reported)
	Wang *et al*, 2012[42]	∝Dominant limb ACLR: 4.4°	14±4 M	6.6°	Described significantly lower internal rotation in ACLR knee (p value not reported)
	Wang *et al*, 2012[42]	∝Non-dominant limb ACLR: 5.2°	14±6 M	6.2°	No differences reported
≥3Y	Webster *et al*, 2012[25]	10.4±4.5°	3±1 Y	12.5±4.4°	−0.46 (−1.16 to 0.24)
Peak external rotation angle					
1 to <3Y	Scanlan *et al*, 2010[28]	3.6°	24 (7–65) M	1.2°	Reported greater external rotation in ACLR knee (p<0.01)
	Wang *et al*, 2012[42]	∝Dominant limb ACLR: 5.0°	14±4 M	4.6°	Described greater external rotation in ACLR knee (p value not reported)
		∝Non-dominant limb ACLR : 4.0°	14±6 M	3.0°	No differences reported

Data are presented as mean±SD.
∝, Lower limb dominance defined by ball kicking; ACLR, anterior cruciate ligament reconstruction; AMP, anteromedial portal; M, month; RTS, return to sport; TT, transtibial; W, week; Y, year.

knee OA populations.[52–54] However, sensitivity analyses revealed strong evidence of lower peak knee adduction angles (ie, less knee varus) in individuals 6–12 months post-ACLR with hamstring-tendon graft compared to healthy controls, but no evidence in those with a patellar-tendon graft. Greater knee abduction angle (equivalent to lower knee adduction) increases the risk of non-traumatic lateral knee OA progression,[55] and thus, if such findings (lower knee adduction angles) persist beyond 6 months, hamstring-tendon ACLR may increase the risk of lateral post-traumatic knee OA. However, this finding should be interpreted with caution, since it unknown whether the cohorts who underwent hamstring-tendon graft ACLR, and were included in this systematic review, had valgus malalignment pre-surgery.

Transverse plane biomechanics data were not well presented in the available literature. Out of the 34 studies included in this systematic review, only nine studies presented transverse plane kinematics data. For four of those nine studies, peak values were estimated from graphs and SDs/ranges were not available. In addition, no available studies presented transverse plane biomechanics data in those who were within 6 months after surgery. Better standardisation of transverse plane biomechanics data in future longitudinal studies will allow enhanced understanding of transverse plane biomechanical features necessary for optimal recovery after surgery, as well as the influence of these biomechanical features on development and progression of post-traumatic knee OA.

It has been theorised that abnormal transverse plane motions following ACL injury alters cartilage load distribution patterns, which may contribute to initiation of early degenerative changes in the joint.[3] Unfortunately, due to the large variability in the reporting of transverse plane kinematics and moments, meta-analysis could not be performed. In those who were 6–12 months postsurgery, there is conflicting evidence provided by single studies as to whether peak knee internal rotation angles are increased or decreased in the first half of stance in individuals following ACLR relative to healthy controls.[36] Shi *et al*,[5] also reported altered transverse plane kinematics in individuals after ACLR. Shi *et al*,[5] reported greater maximum external rotation angle in individuals after ACLR compared to healthy controls, based on pooled data from two studies. While it appears that abnormal transverse plane biomechanics are evident after ACLR, the importance and relevance of such biomechanical alterations are unknown, and should be investigated further.

This systematic review provides evidence that individuals following ACLR have altered gait mechanics, mostly in the sagittal plane, which have the potential to contribute to the development of post-traumatic knee OA. Interventions such as gait retraining or foot orthoses may assist in correcting gait characteristics in individuals following ACLR. However, future

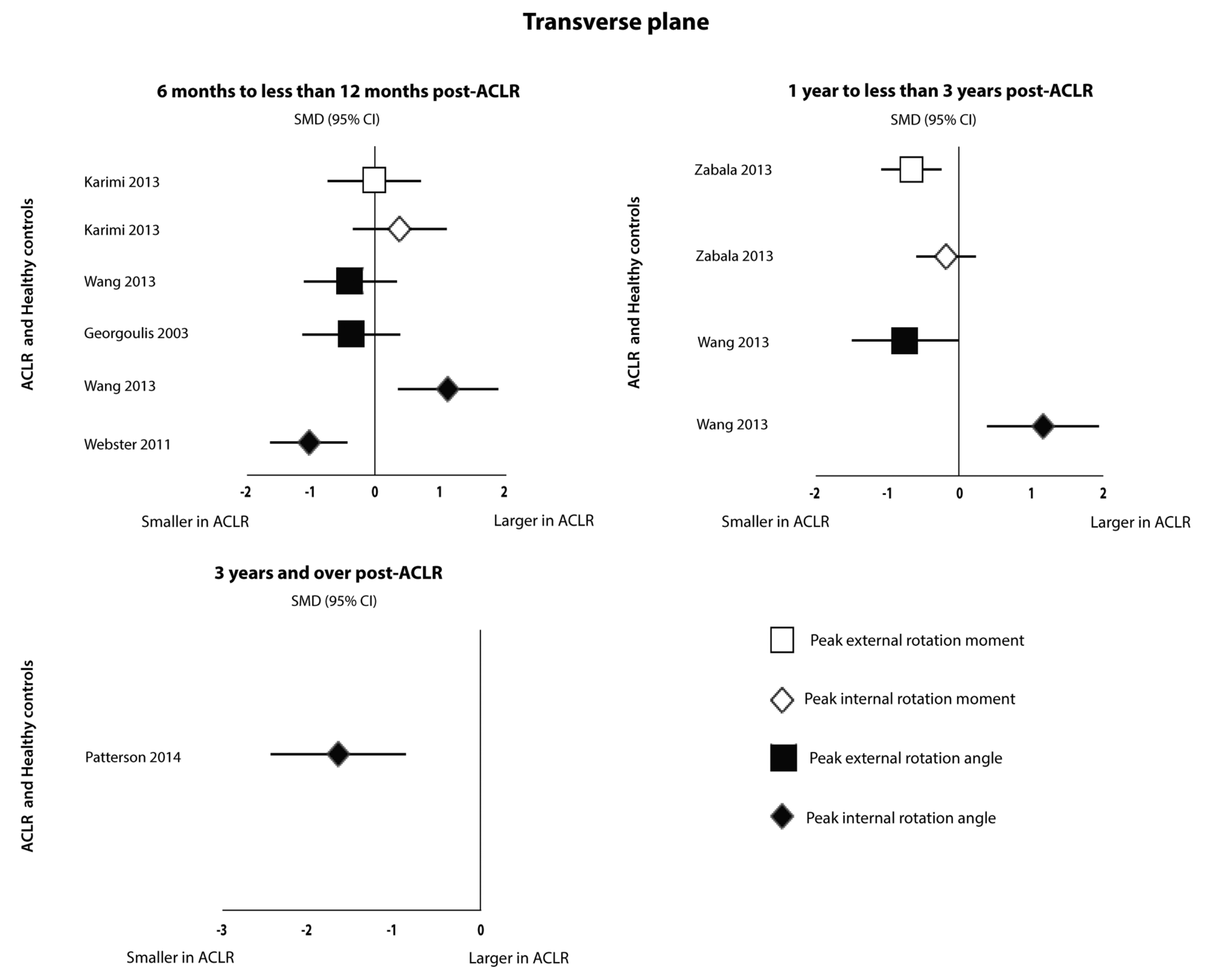

Figure 6 Standardised mean difference (SMD) for peak transverse plane knee joint kinematics and moments. SMD >0: peak knee joint kinematics/moments larger in anterior cruciate ligament reconstruction (ACLR) participants compared to healthy controls. SMD <0: peak knee joint kinematics/moments smaller in ACLR participants compared to healthy controls.

research should consider the timing of measurement since ACLR when investigating potential interventions for this patient population, as biomechanical features noted within 6 months do not appear to match those noted at 12 months postsurgery and beyond. Potential interventions targeted at this patient population may aid in improving knee OA prognosis following ACLR.

Meta-analysis conducted in this systematic review revealed smaller knee flexion angles (after 12 months following surgery) and no differences in knee abduction moment (KAM) in individuals after ACLR compared to healthy controls. Smaller knee flexion angles and larger KAM (ie, smaller KAM) are risk factors of non-contact ACL injury.[56] Considering that not all individuals develop post-traumatic knee OA after ACLR, it is plausible that altered gait mechanics may be evident in ACLR knees before injury or surgery. Thus, longitudinal data (preinjury to postsurgery) will assist in establishing biomechanical risk factors, which may be targeted to ensure optimal recovery after ACLR.

There are number of limitations that should be acknowledged. First, all relevant studies, regardless of methodological quality, were included in this systematic review. Therefore, it is possible that this systematic review is subject to bias through the inclusion of poor quality studies. However, the levels of evidence provide a synthesis of the quality, quantity and homogeneity of study results. Second, due to limited translation resources, we restricted our search to studies published in English. Thus, consideration of data from non-English language studies may alter the outcomes presented in this paper. Third, this systematic review evaluated knee kinematics and moments in individuals following autograft ACLR. Therefore, the results of this systematic review may not be generalised to individuals after allograft ACLR. Finally, a small number of studies contributed to each meta-analysis, and thus results of this systematic review should be interpreted with caution. However, the systematic review and meta-analysis did reveal a paucity of longitudinal data in this patient-population.

In summary, there is moderate evidence indicating that sagittal plane knee angles and moments during walking are altered after ACLR. Notably, there was strong evidence of no difference in knee adduction moment between ACLR and control participants. Sensitivity analyses indicate that hamstring-tendon grafts may be associated with increased knee adduction angle during gait, whereas patellar-tendon grafts are not. Owing to very limited evidence, no conclusions can be drawn regarding transverse plane kinematics. Longitudinal studies are required to identify biomechanical features at the knee and secondary joints that are associated with the development and/or progression of post-traumatic knee OA after ACLR.

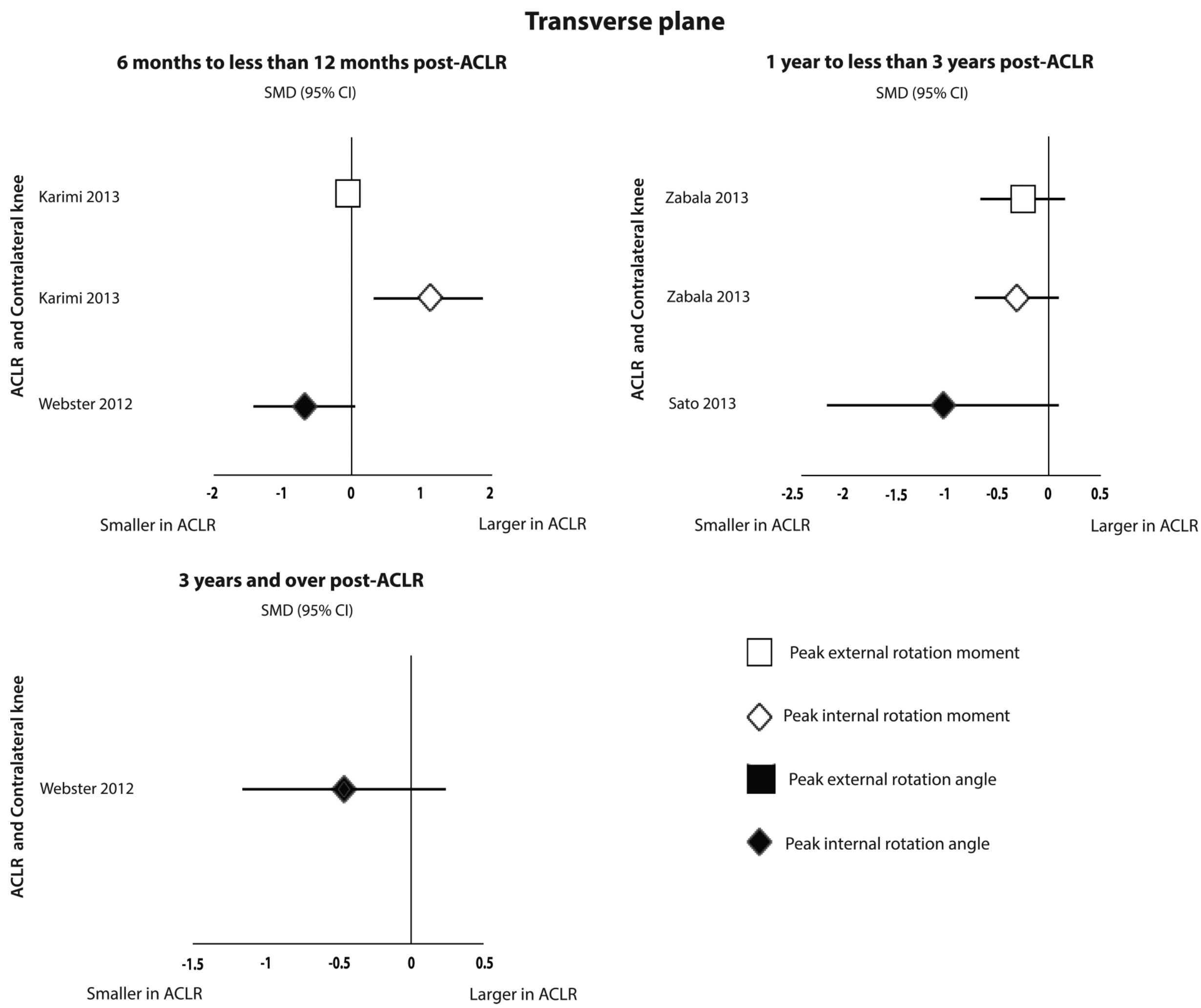

Figure 7 Standardised mean difference (SMD) for peak transverse plane knee joint kinematics and moments. SMD >0: peak knee joint kinematics/moments larger in anterior cruciate ligament reconstruction (ACLR) knees compared to contralateral knees. SMD <0: peak knee joint kinematics/moments smaller in ACLR knees compared to contralateral knees.

Table 7 Transverse plane knee moments

	Author, year	ACLR cohort	Time post-ACLR	Comparator	Standardised mean difference (95%CI)
ACLR vs healthy controls					
Peak internal rotation moment					
6 to <12M	Karimi *et al*, 2013[30]	0.23±0.19 Nm/kg	6 M	0.17±0.09 Nm/kg	0.39 (−0.33 to 1.12)
1 to <3Y	Zabala *et al*, 2013[32]	0.97±0.24 Nm/(%kg m)	26 (22–36) M	1.01±0.24 Nm/(%kg m)	−0.17 (−0.58 to 0.25)
Peak external rotation moment					
6 to <12M	Karimi *et al*, 2013[30]	0.13±0.06 Nm/kg	6 M	0.13±0.06 Nm/kg	0.00 (−0.72 to 0.72)
1 to <3Y	Zabala *et al*, 2013[32]	0.17±0.11 Nm/(%kg m)	26 (22–36) M	0.24±0.10 Nm/(%kg m)	−0.66 (−1.09 to −0.24)
	Bulgheroni *et al*, 1997[58]	16.1 Nm/(%kg m)	17±5 M	6.7 Nm/(%kg m)	Not reported
ACLR vs contralateral knee					
Peak internal rotation moment					
6 to <12M	Karimi *et al*, 2013[30]	0.23±0.19 Nm/kg	6 M	0.08±0.04 Nm/kg	1.06 (0.29 to 1.83)
1 to <3Y	Zabala *et al*, 2013[32]	0.97±0.24 Nm/(%kg m)	26 (22–36) M	1.04±0.21 Nm/(%kg m)	−0.31 (−0.72 to 0.11)
Peak external rotation moment					
6 to <12M	Karimi *et al*, 2013[30]	0.13±0.06 Nm/kg	6 M	0.20±0.19 Nm/kg	−0.07 (−0.17 to 0.03)
1 to <3Y	Zabala *et al*, 2013[32]	0.17±0.11 Nm/(%kg m)	26 (22–36) M	0.20±0.12 Nm/(%kg m)	−0.26 (−0.67 to −0.16)

Data are presented as mean±SD.
ACLR, anterior cruciate ligament reconstruction; kg, kilogram; m, metre; M, month; NM, Newton metre.

What are the new findings?

- Individuals post-anterior cruciate ligament reconstruction (ACLR) have altered sagittal plane kinematics and moments compared to healthy controls and uninjured contralateral knees.
- Knee adduction moments are maintained in individuals post-ACLR compared to healthy controls and contralateral knees.
- Hamstring tendon graft may influence knee adduction angles in individuals post-ACLR.

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Contributors HFH and KMC were involved in the conception and design. HFH and KMC were involved in the search strategy. HFH, AGC, DCA and SMC were involved in the screening of the articles. HFH and NJC were involved in data extraction. ZM, AGC and NJC were involved in the methodological quality ratings. HFH, NJC and KMC were involved in the data analysis.

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